OmniSearch

How HAPPO reached ≥0.9 recall

The kinds of changes that took survivor recall (fraction of 5 survivors confirmed) from a do-nothing **0.0** to a **learned 0.97** — honestly, under realistic sensors. Config: 1 km terrain, floor 0, line-of-sight enforced, drone EO/IR confirmation, `n_ground=2`, 1000-step episodes.

Metric	Value
Learned HAPPO (BC + RL)	**0.97**
Learned HAPPO under 30% comms dropout	0.97
From-scratch RL (plateau)	0.60
Lawnmower expert (ceiling)	1.00

The one-line story: three kinds of change, in order — **fix the world** (make the map small enough that detection is physically possible), **fix the model, not the sensors** (let drones confirm from altitude with line-of-sight — the honest route to ≥0.9), and **fix how we learn** (clone the expert, then RL fine-tune, and keep the best checkpoint).

The recall journey

Stage	Conditions	Recall	Real?
Pure RL (early)	1 km, basic reward shaping	~0.20	real but weak
Generous sensors	confirm range ≈ ½ map, see-through terrain	0.90	**artifact**
+ Line-of-sight gate	same, but confirmation needs a clear view	0.40	exposes the artifact
From-scratch (realistic)	floor 0, LOS, drone-confirm	0.60	real, but below ceiling
Lawnmower expert	floor 0, LOS, drone-confirm	1.00	real (scripted)
BC warm-start + RL fine-tune	floor 0, LOS, drone-confirm	**0.97**	**real (learned)**

Recall = confirmed / 5 survivors, mean over 6 seeds × 1000-step episodes. "Artifact" = the number was real but came from trivializing the task, not from coordination.

Honesty — the 0.90 was an artifact: requiring line-of-sight collapses the generous-sensor 0.90 to 0.40; realistic aerial confirmation recovers it to 1.00 honestly (0.90 → 0.40 → 1.00).

The learned-policy lever: under the realistic config, cloning the expert then RL-fine-tuning lifts the learned policy from 0.60 to 0.97 — close to the scripted 1.00 ceiling.

What kinds of changes — and why each mattered

1. Fix the world geometry — precondition

Switched from the default ~22 km map to a **1 km terrain**. On the big map a survivor is a ~0.04 %-of-map pinpoint and **everyone scored 0.0** — the geometry, not the policy, was the blocker.

Effect: `sim_units_per_meter` ~20× larger → confirmation becomes physically possible. Lifts recall off zero.

2. Fix the model, not the sensors — decisive

Added `drone_can_confirm` — drones confirm survivors from altitude with a clear top-down view (real EO/IR aerial SAR) — and a `confirm_requires_los` line-of-sight gate so nothing confirms through a mountain.

Effect: honest ceiling rises to 0.83–1.00 with realistic FOV/altitude, instead of inflating sensor range. The LOS gate proves the recall is real.

3. Fix how we learn — got us to 0.9

Behaviour-clone the 1.0-recall lawnmower into all 5 HAPPO actors (`train_bc_happo.py`), then **RL fine-tune** from that warm-start (`--model-dir`). Keep the best checkpoint by recall, not the last.

Effect: learned policy 0.60 → 0.97. Fine-tune starts at ~79 reward vs ~42 from scratch; snapshotting avoids the late-run regression (final fell to 0.80).

4. Reward & observation shaping — supporting

Team `coverage_obs_grid` so the policy can learn systematic sweeping, a confirmation-dominant reward, an idle/pending penalty, a ground-exploration reward, and a recurrent **GRU** policy.

Effect: necessary plumbing so the clone matches the obs space and RL can express the sweep-and-confirm behaviour. On their own (from scratch) they reach only ~0.20–0.70.

The two-step path that reached 0.9

Step 1 · Behaviour cloning → Step 2 · RL fine-tune → Learned 0.97

Step 1: roll out the lawnmower expert under the realistic config and clone its actions into the HAPPO actors with recurrent BPTT (NLL 0.08 → −1.3). **Step 2:** load those actors and run HAPPO (critic trains fresh) for 1M steps. Evaluate periodic snapshots and keep the best (`results/happo_realistic_best`, 0.97 recall, robust to 30% comms dropout).

What to claim — and what not to: the credible headline is **≥0.9 by realistic aerial confirmation** (expert 0.83–1.00; learned 0.97). The earlier flat 0.90 from generous sensors + see-through terrain should only ever be cited as a sensor-sensitivity upper bound, never as the operating point.